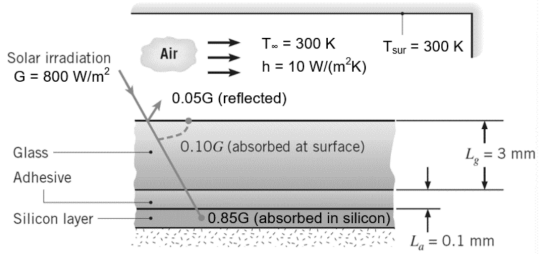
Problem #1

Known: L_g, k_g, L_a, k_a, $\eta = a - bT_{si}$, a, b, R_c, G, $G_{g,top}$(absorbed), $G_{g,top}$(reflect), G_{si}, T_∞, h, T_{sur}, h_{rad}

Find: a) thermal resistance network w/ clear labels and assumptions and each thermal resistance b) $T_{g,top}$ & T_{si} c) η and electricity power flux

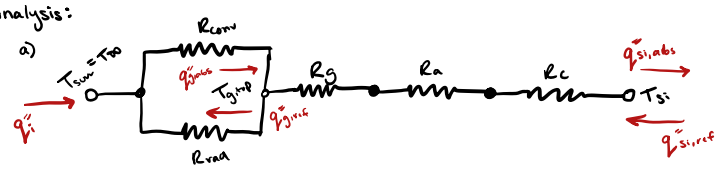
Schematic:



Assumptions: Backside of si layer is thermally insulated, no contact resistance between glass and adhesive, no σ for problem

Analysis:

a)



$R_c = 10^{-4}\ m^2K/W$

$R_a = 1\times10^{-4}/1 = 10^{-4}\ m^2K/W$

$R_g = 3\times10^{-3}/1.5 = 2\times10^{-3}\ m^2K/W$

$R_{conv} = 1/10 = 0.1\ m^2K/W$

$R_{rad} = 1/5 = 0.2\ m^2K/W$

$R_{//} = \frac{1}{5} + \frac{1}{10} = 0.067\ m^2K/W$

$q''_i = G = 800\ W/m^2$

$q''_{g,abs} = G_{g,top}(\text{absorb}) = 0.10\,G = 80\ W/m^2$

$q''_{g,ref} = G_{g,top}(\text{reflect}) = 0.05\,G = 40\ W/m^2$

$q''_{si} = G_{si} = 0.85\,G = 680\ W/m^2$

b)

$$q''_{si,therm} = (1-\eta)(q''_{si}) = (1-\eta)(680\ W/m^2)$$

$$\eta = a - bT_{si} = 0.553 - 0.001\,T_{si}$$

$$(1-(0.553-0.001T_{si}))(680\ W/m^2)$$

$$= (1 - 0.553 + 0.001T_{si})(680)$$

$$= 303.96 + 0.68\,T_{si}$$

$$q''_{si,therm} = \frac{T_{si} - T_{g,top}}{R_a + R_c + R_g}$$

$$(303.96 + 0.68T_{si}) = \frac{T_{si} - T_{g,top}}{0.0022}$$

$$0.669 + 0.0015T_{si} = T_{si} + T_{g,top}$$

$$0.669 - 0.999\,T_{si} = -T_{g,top}$$

$$q''_{si,therm} + 80 = 15(T_{g,top} - 300)$$

$$(303.96 + 0.68T_{si}) + 80 = 15(T_{g,top} - 300)$$

$$20.264 + 0.0453T_{si} + 5.33 = T_{g,top} - 300$$

$$325.594 + 0.0453\,T_{si} = T_{g,top}$$

$$\boxed{\left.\begin{array}{l} T_{si} \approx 342.1\ K \\ T_{g,top} \approx 341.1\ K \end{array}\right\} \text{used Desmo to solve}}$$

c)

$$\eta = 0.553 - 0.001(342.1) = \boxed{0.2109}$$

$$q''_{electricity} = \eta(680) = (0.2109)(680) = \boxed{143.4\ W/m^2}$$